

Hongting Liu

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Professional Summary

Software Engineer with an MSc in Computer Science: Robotics and Computation from University College London and a BSc in Computer Science: Artificial Intelligence. Experienced in back-end systems, database optimisation, AI-driven applications, and real-time interactive systems across Python, SQL, C#, and JavaScript. Strong foundation in distributed systems, machine learning, data analytics, and production engineering, with project experience in AI scene synthesis, audio-based accessibility, and scalable infrastructure.

Education

University College London

Sep 2023 – Jun 2024

MSc Computer Science: Robotics and Computation (Merit)

London, UK

- Key Modules: Distributed Systems, Applied Deep Learning, Data Analytics, Multi-Agent Systems

University of Sussex

Sep 2018 – Jun 2023

BSc Computer Science: Artificial Intelligence (2:1 Honours)

Brighton, UK

- Key Modules: Computer Vision, NLP, Database Systems, Statistics, Machine Learning

Experience

Software Engineer

Feb 2024 – Jun 2024

GroupMe Ltd

London, UK

- Designed and scaled real-time push notification infrastructure supporting thousands of concurrent users, improving reliability and response performance during peak traffic.
- Optimised MySQL queries and back-end architecture, reducing latency by 30% and enabling high-throughput production operations.
- Collaborated with product, UX/UI, and QA teams to ship user-facing features, using monitoring and performance feedback to improve reliability after release.

Back-End Engineer Intern

Jun 2021 – Sep 2021

Guangdong Excellent Gas Group Investment Co., Ltd

Guangdong, China

- Designed scalable back-end services with Python and Node.js, improving data pipeline stability and reducing latency by 25%.
- Deployed Redis caching for high-frequency queries, improving throughput and reducing repeated database access under peak-load scenarios.

Projects

AI Scene Synthesis System

Jun 2024 – Sep 2024

University College London

- Built an AI-driven 3D scene synthesis pipeline using Python and deep learning workflows, processing language-conditioned scene data and improving inference efficiency through batching, preprocessing, and pipeline optimisation.

Sound-Navigation 3D Game

Dec 2022 – Apr 2023

University of Sussex

- Developed a Unity-based 3D accessibility game with real-time audio navigation, implementing low-latency C# feedback loops and evaluating usability through player testing.

Technical Skills

Programming: Python, SQL, JavaScript, C#, Java, Git

Back-End & Infrastructure: Linux, Docker, Kubernetes, Redis, MySQL, REST APIs, Networking

Software Engineering: Distributed Systems, CI/CD, Monitoring, Testing, Incident Response

AI & Data: PyTorch, TensorFlow, OpenCV, scikit-learn, NLP, Data Analytics